Pure Mathematics Year 2

Chapter 4

Binomial Expansion

Negative and fractional powers · validity · approximations · partial fractions

Starters

1. Expand to the term in x^3.

(a) $(1+5x)^7$

1a

(b) $(5-2x)^{10}$

1b

(c) $(1-x)(2+x)^6$

1c

2. Express in partial fractions.

(a) $\dfrac{-14x+7}{(1+2x)(1-5x)}$

2a

(b) $\dfrac{24x-1}{(1+2x)^2}$

2b

(c) $\dfrac{24x^2+48x+24}{(1+x)(3x-4)^2}$

2c

1. New powers

We extend the binomial expansion to negative and fractional powers.

$\sqrt{1+x} = (1+x)^{1/2}$.

The expansion is generally an infinite series, with a restricted interval of validity.

2. A constant other than 1

First factor out the constant:

$$(8+5x)^{-1/3} = \frac{1}{2}\left(1+\frac{5x}{8}\right)^{-1/3}.$$

Then use the binomial formula.

3. Rational expressions

Partial fractions can give simpler expressions to expand:

$$\frac{4-5x}{(1+x)(2-x)}.$$

Expand each fraction, then combine like powers.

Before you start

For every expansion:

1. Identify the power and the expression replacing x.
2. Keep brackets around negative terms.
3. Simplify one coefficient at a time.
4. State the interval of validity.
5. For an approximation, check that the substituted value lies in that interval.

Year 1 recap

Expand each expression.

$(1+x)^5$

1

$(1+2x)^4$

$(1-3x)^3$

Generalised coefficients

$$\binom{n}{r} = \frac{n(n-1)\cdots(n-r+1)}{r!}.$$

This formula also makes sense for non-integer n.

Evaluate $\binom{10}{3}$, $\binom{-1}{2}$, $\binom{-2}{3}$, $\binom{1/2}{2}$.

The binomial series

$$(1+x)^n = 1 + nx + \frac{n(n-1)}{2!}x^2 + \frac{n(n-1)(n-2)}{3!}x^3 + \cdots.$$

For negative or fractional n, use $|x| < 1$. For non-negative integer n, the expansion terminates and is valid for every real x.

Example 1

Expand $(1+x)^{-1}$ to x^3.

1

Example 2

Expand $\sqrt{1-3x}$ to x^3.

When are infinite expansions valid? **Luisa · 7**

Why does validity matter?

$$\frac{1}{1+x} = 1 - x + x^2 - x^3 + \cdots .$$

The right side is a geometric series with ratio $-x$.

For which values of x does it converge?

Validity

Compare these cases

$0 < x < 1$: signs alternate and magnitudes decrease.

$-1 < x < 0$: all terms are positive and decrease.

$x = 1$: partial sums alternate between 1 and 0.

$x > 1$: term magnitudes grow.

$x = -1$: the original fraction is undefined.

Decreasing terms alone do not prove convergence; here the geometric-series result does.

When are infinite expansions valid? **Luisa · 8**

Replace the whole expression

For $\sqrt{1-3x}$, require

$$|-3x| < 1 \iff |x| < \frac{1}{3}.$$

State the standard interval of validity for each expansion.

(a) $(1+2x)^{-1}$

(b) $(1-x)^{-2}$

(c) $(1+x/4)^{1/2}$

(d) $(1-2x/3)^{-1}$

a

b

c

d

Combining Expansions

Edexcel C4 June 2013, Q2

Show that, for small x,

$$\sqrt{\frac{1+x}{1-x}} \approx 1 + x + \frac{1}{2}x^2.$$

Write the expression as $(1+x)^{1/2}(1-x)^{-1/2}$.

Expand

Combine

Test Your Understanding

1. Expand $(1+4x)^{-2}$ to x^3, and state its interval of validity.

1

2. Edexcel C4 January 2010, Q1

(a) Expand $\sqrt{1-8x}$ to x^3.

(b) Show that at $x = 1/100$, the exact value is $\sqrt{23}/5$.

(c) Hence approximate $\sqrt{23}$ to 5 decimal places.

a

Accuracy of an approximation — Luisa · 11

Accuracy of a truncated series

Approximate $1/(1+x)$ using $1 - x + x^2$, with $x = 0.01$.

Approximation

Why is this close?

The first omitted term is $-x^3 = -10^{-6}$, and the later terms are smaller.

Using more terms usually improves an approximation within the interval of convergence.

Values near the boundary can require many more terms for comparable accuracy.

Exercise 4A pages 96-97 — Luisa · 12

Exercise 4A, Question 4

$f(x) = \dfrac{1+x}{1-2x}$.

(a) Find the expansion in ascending powers of x, up to and including x^3.

(b) State the interval of validity.

a

b

Exercise 4A, Question 5

$f(x) = \sqrt{1+3x}, \qquad -\frac{1}{3} < x < \frac{1}{3}.$

(a) Expand to x^3.

(b) Show that $f(0.01) = \sqrt{103}/10$.

(c) Find the percentage error in the series approximation at $x = 0.01$, to 2 significant figures.

a

b

c

Exercise 4A, Question 6

In $(1+ax)^{-1/2}$, the coefficient of x^2 is 24.

(a) Find the possible values of a.

(b) Find the corresponding coefficient of x^3.

6

Question 7

Show that, for small x,

$$\sqrt{\frac{1+x}{1-x}} \approx 1 + x + \frac{1}{2}x^2.$$

7

Exercise 4A, Question 8

$h(x) = \dfrac{6}{1+5x} - \dfrac{4}{1-3x}$.

(a) Expand to x^2. (6 marks)

(b) Find the percentage error when the expansion estimates $h(0.01)$, to 2 significant figures. (3 marks)

(c) Explain why it cannot be used for $h(0.5)$. (1 mark)

a

b

c

Exercise 4A, Question 9

(a) Expand $(1-3x)^{3/2}$ to x^3. (4 marks)

(b) Show that at $x = 1/100$, its exact value is $97\sqrt{97}/1000$. (2 marks)

(c) Hence approximate $\sqrt{97}$ to 5 decimal places. (3 marks)

a

b

c

Factoring Out the Constant — Luisa · 17

When the constant is not 1

Expand $\sqrt{4+x}$ to x^3, and state the interval of validity.

First factor out 4 inside the root:

$\sqrt{4+x} = 2\left(1+\frac{x}{4}\right)^{1/2}$.

Expand

Validity

Quickfire First Step — Luisa · 18

Rewrite first, then state validity.

(a) $(2+x)^{-3}$

a

(b) $(9+2x)^{1/2}$

b

(c) $(8-x)^{1/3}$

c

(d) $(5-2x)^{-3}$

d

(e) $(16+3x)^{-1/2}$

e

Check the prefactor

Apply the power to the factored constant as well: $16^{-1/2} = 1/4$.

Avoid these common errors

Signs

Square the whole term:

$(-3x)^2 = 9x^2, \qquad (-3x)^3 = -27x^3.$

Factorials

The cubic coefficient uses $3! = 6$, not 3.

Brackets

In $(1 - 3x)^n$, every power applies to $-3x$.

Constant factor

In $(a + bx)^n$, first write

$$a^n \left(1 + \frac{bx}{a}\right)^n.$$

Validity

Test $|bx/a| < 1$, not just $|x| < 1$.

Edexcel C4 June 2013 (withdrawn), Q1

(a) Expand $\sqrt{9 + 8x}$, $|x| < 9/8$, up to and including x^2. Give simplified fractional coefficients. (5 marks)

(b) Use the expansion to estimate $\sqrt{11}$ as a single fraction. (3 marks)

a

b

Exercise 4B, Question 2

$f(x) = (5 + 4x)^{-2}, \qquad |x| < \frac{5}{4}.$

Expand in ascending powers of x, up to and including x^3. Give simplified fractional coefficients. (5 marks)

Answer

Exercise 4B, Question 3

$m(x) = \sqrt{4 - x}, \qquad |x| < 4.$

(a) Expand to x^2. (4 marks)

(b) Show that $m(1/9) = \sqrt{35}/3$. (2 marks)

(c) Use (a) to approximate $\sqrt{35}$, and calculate the percentage error. (4 marks)

a

b

c

Exercise 4B, Question 4

The first three terms are

$$\frac{1}{\sqrt{a+bx}} = 3 + \frac{1}{3}x + \frac{1}{18}x^2 + \cdots .$$

(a) Find a and b.

(b) Find the coefficient of x^3.

a

b

Exercise 4B, Question 5

$$f(x) = \frac{3 + 2x - x^2}{4 - x}.$$

Prove that, for sufficiently small x,

$$f(x) \approx \frac{3}{4} + \frac{11}{16}x - \frac{5}{64}x^2.$$

Answer

Exercise 4B

Luisa · 25

AEA 2006, Question 1

(a) Expand $(1-y)^{-2}$ to y^3.

(b) Express the infinite series as $(a+x)^n$:

$$1+\frac{2x}{1+x}+\frac{3x^2}{(1+x)^2}+\cdots+\frac{rx^{r-1}}{(1+x)^{r-1}}+\cdots.$$

(c) State the values of x for which this is valid.

a

b

c

Using Partial Fractions

Luisa · 26

Expand using partial fractions

Expand to x^3 and state validity:

$$\frac{4-5x}{(1+x)(2-x)}.$$

Partial fractions

Expansion

Validity

Test Your Understanding — Luisa · 27

Edexcel C4 June 2010, Q5

$$\frac{2x^2+5x-10}{(x-1)(x+2)} \equiv A + \frac{B}{x-1} + \frac{C}{x+2}.$$

(a) Find A, B, C. (4 marks)

(b) Expand to x^2, with simplified fractional coefficients. (7 marks)

a

b

Exercise 4C Pages 102-103 — Luisa · 28

Exercise 4C, Question 1

(a) Express in partial fractions:

$$\frac{8x+4}{(1-x)(2+x)}.$$

(b) Expand to x^2.

(c) State the interval of validity.

a

b

c

Exercise 4C, Question 2

(a) Express $-2x/(2+x)^2$ in partial fractions.

(b) Determine B, C in its expansion

$-\frac{1}{2}x + Bx^2 + Cx^3 + \cdots$.

(c) State validity.

a

b

c

Exercise 4C, Question 4

$$g(x) = \frac{12x-1}{(1+2x)(1-3x)}, \qquad |x| < \frac{1}{3}.$$

(a) Find A, B such that

$$g(x) = \frac{A}{1+2x} + \frac{B}{1-3x}.$$

(b) Expand $g(x)$ to x^2.

a

b

Exercise 4C, Question 6

$$\frac{3x^2+4x-5}{(x+3)(x-2)} \equiv A + \frac{B}{x+3} + \frac{C}{x-2}.$$

(a) Find A, B, C. (4 marks)

(b) Expand to x^2, giving simplified fractional coefficients. (7 marks)

a

b